Amazon VPC — Important Interview Questions & Answers

1. What is Amazon VPC?
Amazon VPC is a private network in AWS where we can create resources like EC2 and databases.
We can control the IP addresses, subnets, routing, and security of these resources.
Simple ga gurthupettuko:
VPC ante AWS lo mana resources kosam create chesukune private network.
Eng: VPC is a private network created in AWS for our resources.

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2. What is a CIDR Block?
CIDR block defines the IP address range available inside a VPC or subnet.
For example, 10.0.0.0/16 can be used as the CIDR block for a VPC.
Simple ga gurthupettuko:
CIDR ante VPC or subnet lo use cheyyagalige IP address range.
Eng: CIDR defines the IP address range of a VPC or subnet.

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3. What is a Subnet?
A subnet is a smaller network created inside a VPC.
We can create public and private subnets based on our application requirements.
Simple ga gurthupettuko:
Subnet ante VPC lopala create chese smaller network.
Eng: A subnet is a smaller network created inside a VPC.

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4. What is a Public Subnet?
A public subnet is a subnet that has a route to an Internet Gateway.
Resources in this subnet can communicate with the internet when the required settings are configured.
Simple ga gurthupettuko:
Public Subnet ante Internet Gateway ki route unna VPC subnet.
Eng: A public subnet is a subnet that has a route to the Internet Gateway.

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5. What is a Private Subnet?
A private subnet does not have a direct route to an Internet Gateway.
Resources in this subnet can use a NAT Gateway when they need internet access.
Simple ga gurthupettuko:
Private Subnet ante direct Internet Gateway route leni VPC subnet.
Eng: A private subnet does not have a direct route to the Internet Gateway.

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6. What is the difference between Public and Private Subnet?
A public subnet has a route to the Internet Gateway, while a private subnet does not.
Private subnet resources can use a NAT Gateway for outgoing internet access.
Simple ga gurthupettuko:
Public Subnet ki Internet Gateway route untundi, Private Subnet ki direct route undadu.
Eng: A public subnet has an Internet Gateway route, while a private subnet does not.

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7. What is an Internet Gateway?
An Internet Gateway allows resources in a VPC to communicate with the internet.
It is mainly used with public subnets.
Simple ga gurthupettuko:
Internet Gateway ante VPC resources ki Internet tho communication provide chese gateway.
Eng: An Internet Gateway allows VPC resources to communicate with the internet.

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8. What is a Route Table?
A Route Table contains rules that decide where network traffic should go.
It tells the traffic which gateway or resource should receive it.
Simple ga gurthupettuko:
Route Table ante network traffic ekkadiki vellalo decide chese rules.
Eng: A Route Table contains rules that decide where network traffic should go.

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9. What is a Route?
A route tells the network where traffic should go for a particular destination.
For example, 0.0.0.0/0 can send internet traffic to an Internet Gateway.

Simple ga gurthupettuko:
Route ante traffic ye destination ki vellalo cheppe rule.
Eng: A route tells network traffic where it should go.

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10. What is a NAT Gateway?
A NAT Gateway allows resources in a private subnet to access the internet from inside the VPC.
It does not allow the internet to directly start a connection to those private resources.
Simple ga gurthupettuko:
NAT Gateway ante Private Subnet resources ki outgoing Internet access provide chestundi.
Eng: A NAT Gateway provides outgoing internet access to private subnet resources.

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11. Why do we use a NAT Gateway?
We use a NAT Gateway when private resources need to access the internet.
For example, a private EC2 instance can use it to download software updates.
Simple ga gurthupettuko:
Private EC2 ki Internet nundi data download cheyyali ante NAT Gateway use chestham.
Eng: We use a NAT Gateway when private resources need internet access.

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12. What is a Security Group?
A Security Group is a firewall that controls traffic to and from AWS resources like EC2.
We can allow traffic based on ports, protocols, and IP addresses.
Simple ga gurthupettuko:
Security Group ante AWS resource ki traffic ni allow or control chese firewall.
Eng: A Security Group is a firewall that controls traffic to and from a resource.

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13. What is a Network ACL?
A Network ACL is a firewall that controls traffic at the subnet level.
It can have both allow and deny rules for incoming and outgoing traffic.
Simple ga gurthupettuko:
NACL ante subnet level lo traffic ni allow or deny chese firewall.
Eng: A NACL is a firewall that controls traffic at the subnet level.

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14. What is the difference between Security Group and NACL?
Security Group works at the resource level and is stateful.
NACL works at the subnet level and is stateless.
Simple ga gurthupettuko:
Security Group resource level lo work chestundi, NACL subnet level lo work chestundi.
Eng: A Security Group works at the resource level, while a NACL works at the subnet level.

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15. What is a VPC Peering Connection?
VPC Peering allows two VPCs to communicate with each other using private IP addresses.
The two VPCs must have non-overlapping IP ranges.
Simple ga gurthupettuko:
VPC Peering ante two VPCs madhya private communication create cheyyadam.
Eng: VPC Peering allows two VPCs to communicate privately.

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16. Can two VPCs with the same CIDR be connected using VPC Peering?
No, VPC Peering does not support overlapping CIDR ranges.
Each VPC should have a different IP address range.
Simple ga gurthupettuko:
VPC Peering cheyyali ante rendu VPCs CIDR ranges overlap avvakudadhu.
Eng: Two VPCs cannot be peered if their CIDR ranges overlap.

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17. What is a VPC Endpoint?
A VPC Endpoint allows resources in a VPC to connect to supported AWS services without using the public internet.
It is useful for securely accessing AWS services from private subnets.
Simple ga gurthupettuko:
VPC Endpoint ante VPC nundi AWS services ki private connection provide chestundi.
Eng: A VPC Endpoint provides private access from a VPC to supported AWS services.

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18. What is a Gateway Endpoint?

A Gateway Endpoint provides private access from a VPC to services such as Amazon S3 and DynamoDB.
It does not require a NAT Gateway for this service traffic.
Simple ga gurthupettuko:
Gateway Endpoint ante mainly S3 and DynamoDB ki private access kosam use chestham.
Eng: A Gateway Endpoint provides private access to services like S3 and DynamoDB.

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19. What is an Interface Endpoint?
An Interface Endpoint provides private access to supported AWS services using network interfaces inside the VPC.
It uses AWS PrivateLink technology.
Simple ga gurthupettuko:
Interface Endpoint ante VPC lo network interface use chesi AWS services ki private access ivvadam.
Eng: An Interface Endpoint provides private access using network interfaces in the VPC.

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20. What is a Default VPC?
A Default VPC is automatically created by AWS in a Region.
It provides basic networking resources so we can quickly launch AWS resources.
Simple ga gurthupettuko:
Default VPC ante AWS automatic ga create chese ready-to-use VPC.
Eng: A Default VPC is a ready-to-use VPC created automatically by AWS.

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21. What is a Custom VPC?
A Custom VPC is a VPC that we create based on our application requirements.
We can choose its CIDR, subnets, routes, and security settings.
Simple ga gurthupettuko:
Custom VPC ante mana application requirement prakaram maname create chesukune VPC.
Eng: A Custom VPC is a VPC created according to our application requirements.

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22. What is an Elastic Network Interface (ENI)?
An ENI is a virtual network interface that can be attached to an AWS resource.
It can have private IP addresses, security groups, and other network settings.
Simple ga gurthupettuko:
ENI ante AWS resource ki attach ayye virtual network interface.
Eng: An ENI is a virtual network interface attached to an AWS resource.

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23. What are VPC Flow Logs?
VPC Flow Logs record information about network traffic going to and from resources in a VPC.
They are useful for finding network and security problems.
Simple ga gurthupettuko:
VPC Flow Logs ante network traffic ni check chesi problems find cheyyadaniki use chestham.
Eng: VPC Flow Logs record network traffic information for troubleshooting.

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24. What is VPC DNS?
VPC DNS helps resources inside the VPC resolve domain names into IP addresses.
It allows applications and AWS resources to communicate using DNS names.
Simple ga gurthupettuko:
VPC DNS ante resources names ni DNS dwara resolve cheyyadaniki help chestundi.
Eng: VPC DNS helps resources resolve domain names into IP addresses.

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25. Can a VPC have multiple subnets?
Yes, a VPC can have multiple subnets.
We can create these subnets across different Availability Zones for better availability.
Simple ga gurthupettuko:
One VPC lo multiple subnets ni different AZs lo create cheyyachu.
Eng: A VPC can have multiple subnets across different Availability Zones.

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26. Can a subnet belong to multiple VPCs?
No, a subnet belongs to only one VPC.
When we create a subnet, we must select the VPC where it will exist.
Simple ga gurthupettuko:
Oka subnet only one VPC ki matrame belong avutundi.
Eng: A subnet can belong to only one VPC.

27. Can a subnet span multiple Availability Zones?
No, a subnet exists in only one Availability Zone.
For high availability, we create separate subnets in different Availability Zones.
Simple ga gurthupettuko:
Oka subnet only one AZ lo untundi; multiple AZs kosam separate subnets create chestham.
Eng: A subnet exists in only one Availability Zone.

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28. How does a Public EC2 access the internet?
The public subnet needs a route to the Internet Gateway, and the EC2 needs a public IPv4 address.
The Security Group and NACL must also allow the required traffic.
Simple ga gurthupettuko:
Public EC2 Internet ki vellali ante Public IP + IGW route + security rules kavali.
Eng: A public EC2 needs a public IP, Internet Gateway route, and correct security rules.

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29. How does a Private EC2 access the internet?
The private subnet sends internet traffic to a NAT Gateway.
The NAT Gateway then sends the traffic through the Internet Gateway.
Simple ga gurthupettuko:
Private EC2 → NAT Gateway → Internet Gateway → Internet ki vellagaladu.
Eng: A private EC2 can access the internet through a NAT Gateway and Internet Gateway.

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30. Can Private Subnet access Public Subnet?
Yes, resources in private and public subnets can communicate when the route and security rules allow it.
They can communicate using their private IP addresses.
Simple ga gurthupettuko:
Private and Public Subnets madhya communication possible, correct routes and security rules unte.
Eng: Private and public subnets can communicate when routing and security rules allow it.

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31. What is the difference between VPC and Subnet?
VPC is the complete private network in AWS, while a subnet is a smaller network inside the VPC.
One VPC can contain multiple subnets.
Simple ga gurthupettuko:
VPC ante complete network, Subnet ante VPC lopala smaller network.
Eng: A VPC is the complete network, while a subnet is a smaller network inside it.

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32. How do you create a Public Subnet?
First, we create a subnet inside the VPC and create a route to the Internet Gateway.
Then we associate the subnet with that route table.
Simple ga gurthupettuko:
Public Subnet create cheyyali ante subnet ki Internet Gateway route ivvali.
Eng: A public subnet needs a route to the Internet Gateway.

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33. How do you create a Private Subnet?
We create a subnet inside the VPC without a direct route to the Internet Gateway.
If internet access is required, we add a route to a NAT Gateway.
Simple ga gurthupettuko:
Private Subnet ki direct IGW route undadu; Internet kavali ante NAT Gateway use chestham.
Eng: A private subnet has no direct Internet Gateway route and can use a NAT Gateway for internet access.

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34. How would you troubleshoot an EC2 with no internet access?
I would check the subnet route table, Internet Gateway or NAT Gateway, public IP, Security Group, and NACL.
I would also check whether the required route and network settings are correct.
Simple ga gurthupettuko:
EC2 Internet problem vaste Route Table → IGW/NAT → Public IP → SG → NACL check chestham.
Eng: For an internet problem, check the route table, gateway, IP, Security Group, and NACL.

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35. How would you troubleshoot a Private EC2 with no internet access?
I would check whether the private subnet has a route to the NAT Gateway.
Then I would check whether the NAT Gateway is in a public subnet with a route to the Internet Gateway.

Simple ga gurthupettuko:
Private EC2 problem vaste NAT Gateway route and NAT public subnet route check chestham.
Eng: For a private EC2 problem, check the NAT Gateway route and its public subnet route.

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36. What is the most important thing to remember about VPC?
VPC is the main networking service used to create and control networks for AWS resources.
The most important concepts are CIDR, subnets, route tables, gateways, Security Groups, NACLs, and VPC Endpoints.
Simple ga gurthupettuko:
VPC ante AWS lo complete networking setup ni create and control cheyyadaniki use chese service.
Eng: VPC is used to create and control the network for AWS resources.